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Machine Learning as a UX Design Material: How Can We Imagine Beyond Automation, Recommenders, and Reminders?

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Abstract

I design user experiences for machine learning (ML) applications across several application domains, from clinical decision support systems to context-aware mobile services, to autonomous cars. In this paper, I share three cases in which I attempted to leverage ML as a design material, envisioning new forms and new purposes for this technology. I reflect on the challenges encountered and the lessons learned. On reflection, I realized that many of the challenges are not unique to particular ML problems or designers, but in the inherent tension between user-centered design and data-driven design. I hope to initiate a reflective discussion on these overarching challenges in designing ML and to highlight the opportunities in addressing them systematically.

Introduction

Machine learning (ML) increasingly plays an important role in shaping how users interact and experience technology products. From mundane spam filters to personalized news feeds to conversational agents, it can seem like ML is in almost every new technology product and service. Both UX practitioners and researchers have noted this trend and have become especially interested in design opportunities surrounding ML.

However, recent studies show that UX practitioners seem unprepared to effectively leverage ML capabilities. They struggle to understand the capabilities and limitations of ML in the context of their designs, even though many of them understand how ML works generally. Also, they typically join projects near the end, after the functional decisions have been made. "*Design teams are simply putting lipstick on the pig*" (Dove et al. 2017). Researchers noted two symptoms of these struggles of designing ML: One is that designers often fail to notice obvious places where ML could improve UX (Yang et al. 2016b). The other is a lack of design innovation in ML: "*Designers often have cliched understandings of this medium, driven by the hype or criticism surrounding the field. For example, some typical stereotypes are that wearables are for fitness, artificial intelligence is for automating tasks.*" (Allen and Hooker 2017)

Recently, HCI/design researchers began taking actions to address this problem. Some focused on teaching the technical concepts of ML to designers (Hebron 2016). Others organized workshops, bringing together groups of artists, designers, and technologists to collectively explore how ML might function as a creative material (Gillies et al. 2016; Kuniavsky, Churchill, and Steenson 2017). This emergent strand of work within HCI has spawned the notion of *ML as a design material* (Dove et al. 2017; Yang et al. 2016b).

My work is a part of this growing area of inquiry. We designed ML systems across several application domains, ranging from clinical decision support systems, to adaptive mobile user interfaces (UIs), to driving style of autonomous cars. These experience have led us to consider the unique challenges of designing ML applications, especially when new designs arose as a part of our user-centered design intention, rather than from an available dataset. To advance on these challenges, we have since worked to better understand ML as a design material. We interviewed experienced designers who have been designing ML systems for more than a decade, to probe the current best practice in the industry (Yang et al. 2018). We analyzed more than 2,400 HCI research publications, mapping out the design opportunities ML's technical advances have made available (Yang, Banovic, and Zimmerman 2018).

In this paper, I share three cases in which I attempted to leverage ML as a design material across multiple application domains. I reflect on the challenges encountered and the lessons learned. My goal is to demonstrate that many of the challenges are not unique to particular ML problems or designers, but are a result of the inherent tension between user-centered design and data-driven design. Through this paper, I hope to initiate an reflective discussion on these overarching challenges in designing ML, and to highlight the opportunities in addressing them systematically.

Machine Learning as a Design Material

While most recent papers discuss challenges, let me start by articulating this vision of *ML as a design material*.

Technology as Design Material

When I am talking about *ML as a design material*, I am talking about its design innovation. Louridas notes the difference between technical and design innovation when describing the difference between designers and engineers. He notes that engineers create new technology that allow new

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capabilities. Designers, he claims, do not invent new technologies. Instead, they create novel and valuable assemblies of known technologies (Louridas 1999).

When taking a technology as a design material, designers first develop a *tacit* understanding of how the technology opens up and constraint design possibilities. They then innovate by engaging in reflective conversations with design materials, envisioning things that have never before existed (Schön 1984; Wiberg 2014). Schön describes how designers reflect in action, how they conceive of what they want to make while in the act of making it (Schön and Bennett 1996). Gaining a deep understanding of how designers engage in a "conversation with their materials" and how the materials "talk back to the designer" is an important component in understanding design and design practice, and an important means of sharing design knowledge across disciplines (Zimmerman, Stolterman, and Forlizzi 2010).

Machine Learning as Design Material: A Vision

The last few years witnessed explosive technical advances in ML. Our review of literature shows that a total of 1,939 HCI publications mentioned "machine learning" since the publication of the first mention in 1969; More than half of the papers addressing ML were published in the past five years. The overwhelming majority of papers appeared in venues with a technical focus (Yang, Banovic, and Zimmerman 2018). This fast growing set of technical capabilities offer exciting new materials for designers.

Technical advances are often followed by design innovation, where designers envision many new product forms that apply the technical advance into different aspects of peoples lives.

For example, the 1962 cassette record represented a technical advance, providing an easier way to make audio recordings. This technical advance was followed by a wave of design innovation. Designers created new forms including home stereo systems, boom boxes, personal players such as the Walkman, automotive tape players, duo-cassette player-recorders that better supported making mix tapes, dictation devices, and phone answering machines.

Another example is haptics. Moussette has an interest in innovating haptics. He sketched with this technology, setting a goal for himself to make a new haptic device each day for several weeks. His work then produced a new language for talking about the aesthetics of haptics, as well as several simple devices that expose haptics' design possibilities beyond a buzzing phone (Moussette 2012).

We have not yet seen a similar type of design innovation taking place with ML; where a technical advance is followed by many new product forms. "*Today, it seems that ML systems are as creative and interesting as the data scientists that make them*." (Dove et al. 2017) This marks a ready opportunity for design innovation. By engaging with and understanding this technology and its continued advances, designers can envision new forms and new purposes for this technology, and radically re-imagine what it might be or might do.

Motivated by this vision, researchers and designers have began to investigate *ML as a design material*. This strand of work is quite different from technical HCI research which typically utilizes ML as a tool to extend or accelerate well established interaction forms. This is also quite different from other lines of design inquiry associated with ML in which designers often join ML development after the functional decisions have been made.

Case Studies

As part of the growing area of inquiry on *ML as a design material*, my work envisions new products that fit ML's technical advances into different aspects of peoples lives. Of these I draw three cases; I reflect on the challenges encountered and discuss how each of the cases advanced my understanding of ML from a UX design perspective.

Adding Machine Learning to Existing UX Design

The first case is drawn on a simple project in which we extend an existing mobile application with a ML feature that reduces navigation and selection efforts.

> *My collaborators and I attempted to enhance our mobile transit app, Tiramisu, with ML-powered adaptive interfaces. The new ML feature is intended to anticipate users menu-item selections, reducing their navigation and selection efforts. Yet we encountered two problems that made this impossible. First, we had not logged the information needed to infer what users most likely wanted to do. Second, we had not properly motivated users to provide good labels that would support adaptation.*
>
> *On reflection, we realized UX designers should identify and refine UI adaptions when sketching wireframes. Although challenging, recognizing learning opportunities when wireframing will be an important goal for future IxD practices.*
>
> *To advance on this challenge, we developed a set of UI design patterns that define where and how ML adaptations might be applied. We also created a new form of wireframe that support UIs that change over time, across use contexts. On this modified wireframe, designers can annotate pre-adapted and adapted interaction paths, and document the data needed to make an inference, a recovery method in the case of an inference error, and an inference quality needed to trigger adaptation.* (Yang et al. 2016b)

This project's design problem, adaptive UI, and its underlying ML model are simplistic in comparison to the ML problems HCI research focuses on today. Yet we encountered unanticipated barriers in this simple project. This is because we had not yet taken ML into consideration at the early stages of the design process.

We learned two lessons. First, understanding how ML works does not sensitize designers to the opportunities to apply ML in their designs. Most researchers on the above project had many years of experience designing, developing, and evaluating ML systems. Yet we still failed to anticipate simple ML opportunities in our design. This is partially

because ML is not part of a typical user-centered design process; wireframing tools or patterns do not yet support the UIs that change over time or personalize to users.

There is a clear need for design research that helps expand designers perceived application of ML, and sensitizes them to the breadth of its design possibilities. In our conversations and workshops with UX practitioners, we noticed that they often failed to recognize many ML applications that they use every day, especially the ones that have so successfully adapted to their interactions that they have become invisible or unremarkable. Instead, designers only referenced a few classic, and somewhat failed, designs exemplars (i.e. Clippy, Tay, email spam filter) (Yang et al. 2018). Design researchers could create sensitizing concepts to communicate ML design opportunities that are not instantly recognizable.

The second lesson is that UX cannot not be an afterthought of ML. The lack of planning for ML could stand as a barrier to operationalizing this technology in current product development processes. The software development community has learned over many years that usability and user experience should be considered early in the development process and not added as an afterthought at the end. HCI has since developed best practices and tools that scaffold better software development processes. We see a similar need in scaffolding ML considerations in UX design.

In our conversation with designers who regularly work on ML systems, we noticed that they "plan for ML" by working with telemetry data on day-to-day basis. In doing so, they searched for user behaviour patterns that ML can potentially leverage (Yang et al. 2018). Many taught themselves to create data visualization tools to capture "rich and compelling user stories", such that the quantitative analysis of user behavior "*do not privilege data scientists*". The designers thus were able to raise new ideas in enhancing their product with ML.

Adding UX to Existing Machine Learning System

The second case is a project in which we design a clinical ML-driven decision support system by applying a classic user-centered design (UCD) workflow (Council 2005).

> *A few years ago, a team of Bioengineering researchers approached us to design a ML system that helps cardiologists better decide whether and when to implant an end-stage heart failure patient with a mechanical heart pump. Like almost all other clinical decision support tools, the system at the time took a prototypical form: It takes in a list of patient condition measures and produces an individualized prediction of patient trajectory, including likely survival and other post-surgical risks.*
>
> *Taking UCD as our starting point, we first interviewed and observed clinicians caring for candidate patients at three different implant centers. Interestingly, our findings revealed that for most cases, clinicians do not find the implant decisions challenging, and thus would not likely engage with a ML system to aid with the decision. Instead, they would value the support for emergent cases, when they have very little data to predict how a critical patient might respond to available therapies. In addition, the implant clinicians would value a ML system that help upstream clinics and hospitals to refer patients to implant centers in time, before patients became too sick for an implant surgery.* (Yang et al. 2016a)
>
> *Since then, we have been working to develop a new ML system that infers implant outcomes of critical patients in intensive care units (ICU). We also have worked to obtain electronic medical records from local and primary care healthcare systems, in order to infer optimal referral windows for each patient. However, the training datasets for such systems are in their nature extremely difficult to curate and label. We do not yet know whether these two design solutions are technically achievable.*

This case illustrates a classic UCD process, from field study to sketching to prototyping (Council 2005). It was successful by many measures, especially through the lens of taking ML as a design material: We actively examined user needs, wants and social contexts; We envisioned new designs that seem technologically feasible, and users are likely to find valuable. However, we encountered significant barriers in prototyping and implementation. The technical and cultural validity of our designs remained unknown until the ML system was built and implemented.

It seems that ML challenges the general idea of prototyping; of making just enough of a system to assess if this is the right direction to go. ML seems to require a much higher level of commitment, requiring an unwieldy amount of data to create a functional prototype. This could conflict with UX mantras like "fail fast, fail often." (Dove et al. 2017) Consequently, in research, it is difficult to experiment with many different design solutions in searching for the best. In the industry, designers were unable to demonstrate or validate the value in their designs through a working prototype as they traditionally did. As a result, they found it difficult to convince leadership to commit to their more innovative designs. They thus often quickly resorted to familiar designs of ML, such as recommenders and reminders (Yang et al. 2018).

How should we sketch and prototype when the design materials in use – large datasets, computational power, time and efforts of data scientists – are costly? Despite efforts to make "ML available to everyone", most designers will face the reality that sufficient data and proficient data scientists are scarce in their teams. A reflective discussion on the current workflow, and how it might adapt to costly ML systems is necessary.

There is also a real need for design tools and methodologies that support designers who lack constant access to capable data scientists. For example, ML tools for designers could simulate the role of the data scientists, enabling designers to quickly evaluate the feasibility of their ideas when sketching. We also see opportunities for research to demonstrate creative designs that use readily available ML solutions (i.e. off-the-shelf ML plugins); designs that do not need intensive ML development effort to implement.

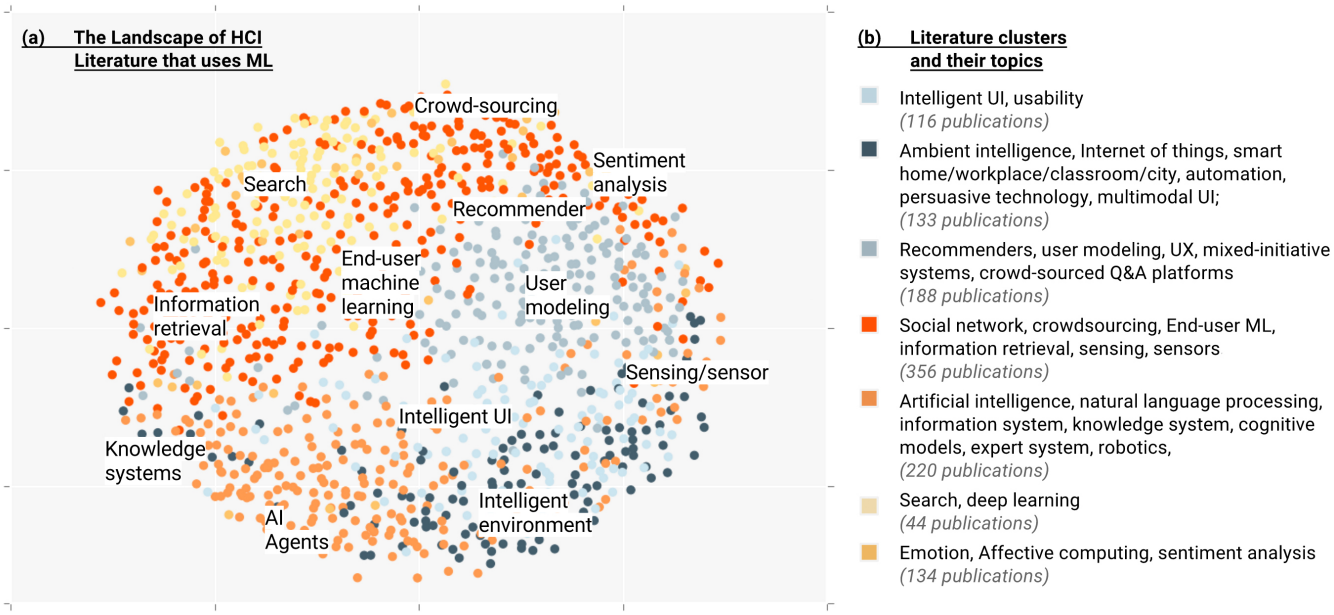

Figure 1: An overview of HCI research that uses machine learning (Yang, Banovic, and Zimmerman 2018). (a) An illustration of the literature landscape based on the semantic distances among each publication. Each dot represents a publication, color-coded by cluster. (b) major topics of each cluster. Topic co-occurrences in these clusters surfaced some common combinations of ML techniques and interaction forms.

UX and ML Match-Making

This leads to the third case in which we worked to help designers identify opportunities in available datasets and algorithms as well as to search for valuable new things to make.

With regard to "technology in searching for users", HCI researchers have previously proposed the *matchmaking* method (Bly and Churchill 1999). Matchmaking starts by asking designers to detail the technical *capabilities* of the tech they are working on. Next, they systematically work to discover *activities* related to these capabilities, *domains* related to the activities, and finally target *users* connected to the revealed domains. Unfortunately, this approach is both under-investigated as a design method and underutilized by both design researchers and practitioners.

We wanted to reveal the technical capabilities and interaction forms HCI has worked on with respect to ML. We wanted to identify how ML has been used to generate or augment value for users. Towards these goals, we analyzed 2,494 HCI publications that mention ML with a combination of manual and algorithmic methods. This process produced three representations of ML's design space (Yang, Banovic, and Zimmerman 2018):

- *7 clusters of work where HCI researchers have repeatedly employed ML techniques to make an advance (Figure 1).*
- *A schema of machine learning capabilities in terms of enhancing sensing, inference and actuation. Generally, these capabilities increase the value of noisy, real-world data, escalating them to higher-level, more meaningful information.*
- *4 value channels through which ML advances provide experiential value for users. For example, ML provides inferences about what might be optimal (e.g. when designing MOOC platforms, ML models ideal students' engagement pattern and informs interaction design).*

The technically defined clusters, capabilities and user value challenges provide starting places to generate unseen designs of ML. Collectively, use of the cluster and model should help design researchers ideate many possible sensitizing concepts. We recommend a process of selecting a technology and then systematically generating ideas from each of the four value channels. This is one form of match-making.

Particularly, three clusters of ML technical advances have not yet been bound to particular utilities, interactions or user experiences. These design-wise underutilized clusters are: deep learning, sentiment analysis and social network mining. Designers and researchers can pick one technology material, match it with diverse interaction forms, and generate many possible ways of leveraging it to provide value to users. For example, what can we design when one is able to capture societal happiness on twitter? What values can we provide for users with deep learning, beyond providing more targeted ads and recognizing objects in an image?

The clusters, schema and value model in this case are

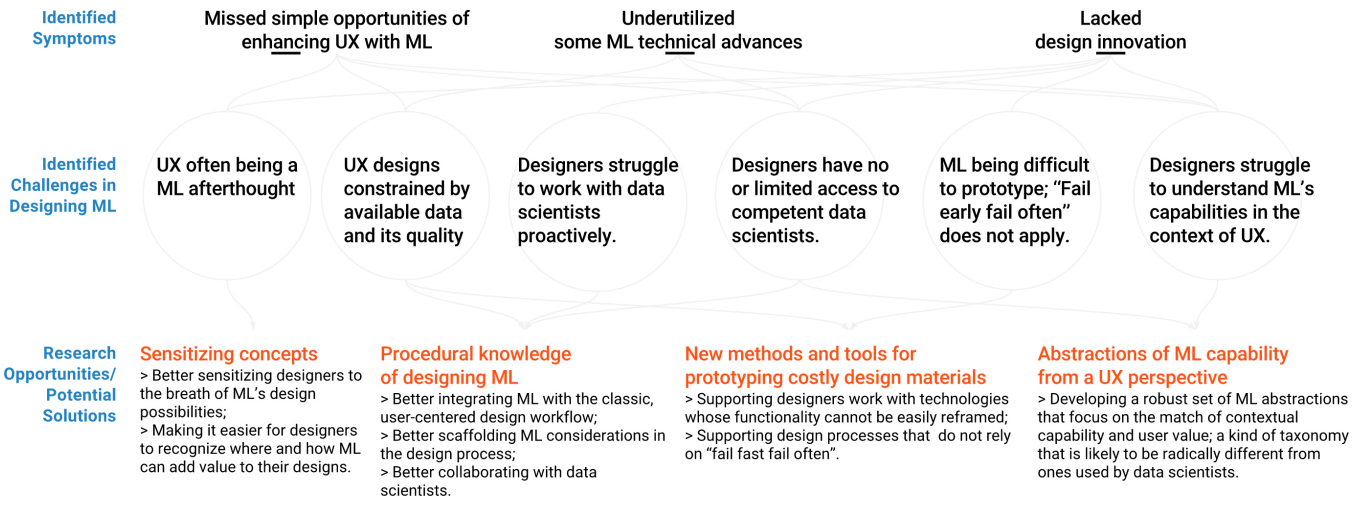

Figure 2: An overview of the challenges in designing ML and the lessons learned in the case studies. These are likely to generalize across many types of ML applications and domains; There is value in investigating these overarching challenges and addressing them systematically.

neither fixed nor final. They await further examination and discussion. Nonetheless, the exploratory process of categorizing ML's technical capabilities and design characteristics was highly informative to us.

We realized that it seems impossible to categorize ML as one design material, and to articulate the capabilities it has or the user activities it can facilitate. For traditional technology materials, for example Bluetooth, researchers provided comprehensive descriptions of its special properties, and then progressed to identify many domains and users these properties might support (Wiberg 2014). In contrast, the capabilities of ML are wedded to its dataset, labels, and underlying algorithm. Its experiential value arises from users' holistic experience over a larger course of interactions. ML in the context of UX design resists easy assimilation into a complete or fixed taxonomy of descriptive mechanisms, such as supervised or unsupervised learning.

Designers need a new way of grasping ML's capabilities, a kind of abstraction that focuses on the match of contextual capability and user value; a kind of taxonomy that is likely to be radically different from ones used by data scientists. The 7 clusters and schema described in the above case provide a starting point for this effort (Figure 1). Additionally, some designers have over the years developed their original abstractions of ML, for example, ML enables *an experience personalized for everyone*", *an evolving relationship with the users*", and "*handling more abstract user instructions*" (Yang et al. 2018). Synthesizing and developing a robust set of such designerly abstractions would help to evolve the understanding of ML as a design material, and to more effectively explore its design possibilities.

Synthesizing the Challenges

The above case studies have illustrated many challenges in innovating ML as a design material as well as the lessons we learned along the way. On reflection, these challenges are not unique to a particular ML problem or application domain, but lie inherently in the characteristics of this design material.

My goal here is not to enumerate the overarching challenges in designing ML, but to highlight the opportunities in addressing them systematically. This section is intended to provide another start point for this emerging area of inquiry (Figure 2).

Inserting ML to Design Practice and Workflow The tension between the classic UCD practice and the designs that arise from available ML systems is a major theme throughout the case studies.

- How to scaffold the design process such that necessary ML considerations can be taken into account timely?
- How to evaluate the technical feasibility of a UX design, especially when proficient data scientists are not constantly accessible?

Without taking ML into consideration throughout product development, even simple designs such as UI adaption could fail. The case studies in this paper demonstrated three kind of workflows; each entailed distinct challenges. Future work should reflect on and improve the workflows we experimented, in searching for better ways of inserting ML to product and service design practice.

At a higher level, the procedural knowledge of designing ML marks a clear space for UX/ML research. Existing work has offered valuable declarative knowledge and conceptual understandings of ML from a design perspective (i.e. algorithm trust, intelligibility, embodiment). Embedding this growing body of new knowledge into organizational and procedural contexts opens up new research opportunities and promises real impact on UX practice.

Sensitizing Designers to Existing ML Design Opportunities Sensitizing UX practitioners to the design opportunities in new technologies is a recurring theme in some

HCI research. In previous cases, such as haptics, researchers demonstrate working examplars of the technology to communicate the design possibilities. Interestingly, demonstrating a functional ML system to designers was not enough to sensitize them, as many of these systems have weak connections to divergent user experiences after repeated use.

- How to recognize and draw inspirations from existing design examplars, given that successful ML interactions are often invisible or unremarkable?
- How to recognize ML opportunities with in our own designs, especially when telemetry data are not readily available?

There is a real need to create sensitizing concepts (Zimmerman, Stolterman, and Forlizzi 2010) that communicate ML design opportunities that are not instantly recognizable. Some work is needed to explore the diverse forms of research artifacts and knowledge representations, and to deliberately choose the ones that most effectively sensitize practitioners.

Developing a Designerly Understanding of ML In order to push the boundaries of what ML might be and might do, we need to first bring clarity to its existing design space and to identify major unknown topics as a basis for future research endeavor. The review of HCI/ML literature is a first step towards this goal (Figure 1).

- How to abstract ML's capabilities from a UX perspective?
- How to sketch and evaluate multiple designs when "fail fast, fail often" is practically impossible in prototyping ML systems?

In addition, the case studies suggested an alternative view to the common assumption that teaching designers how ML works as the most effective way of helping them engage with it as a design material. In match-making ML capabilities and UX possibilities, designers comprehend ML in notably different ways than its textbook definitions. Design researchers hoping to aid practitioners might focus on providing designerly abstractions, exemplars, and new tools that help designers grasp the ML's design potential, quickly sketch and prototype, as well as better collaborate with data scientists. We strongly encourage the UX and HCI research community to join us and start a reflective discussion around innovating ML as a design material.

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